

Lab 10

Modeling Climate Change

now we can swim any day in november

Equipment List

- Computer (lab-provided or your own)

10.1 Introduction

Today we will explore an online climate simulation tool, called **En-ROADS**¹, created by MIT Sloane School of Management and the “environmental think tank” *Climate Interactive*. You should have watched their 10 min “Overview and Introduction” video² already, but you could refresh yourself now with the 2 min “Overview” video at *Help/Video Tools/EN-ROADS Overview*³. Your reading this week from Dessler (Chapter 8: *Predictions of Future Climate Change*) will also help you to understand the definitions and methods of quantities used in this modeling tool.

To get started, take 10–15 min to play with the tool, exploring the options for adjusting the parameters (both using the adjustable bars, and opening the setting to manually control the numbers), visualizing different output graphs, and determining outcomes that differ from the “baseline” presented. Talk with the lab partners at your table about what you are finding and share methods on how to access all of the tools/options. While/after doing so, write a brief paragraph that answers most of the following questions.

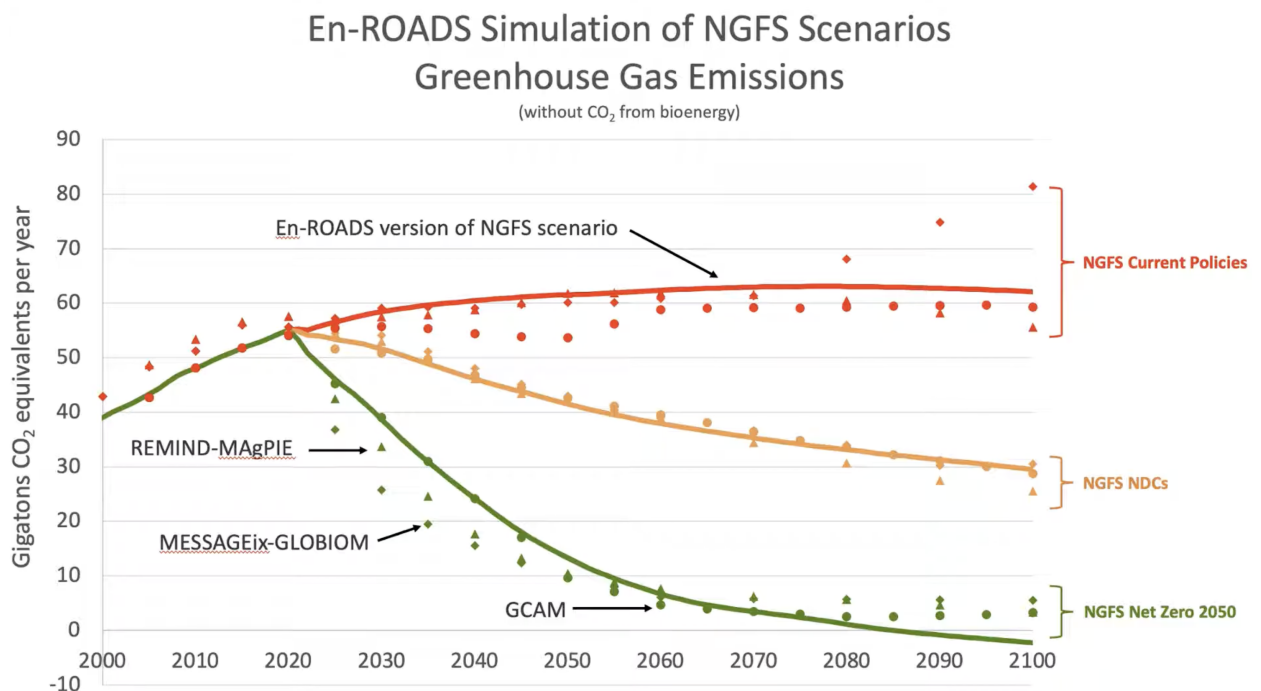
- What is the basic idea of this tool? What are the inputs and what are the outputs?
- Where are the connections to our prior study of climate change, e.g., “climate sensitivity”, the connection between increasing CO₂ in the atmosphere and temperature rise?
- What else is incorporated in the tool, other than these climate physics principles? (I would say it accounts for the science, but also societal and economic factors. How?)

¹<https://en-roads.climateinteractive.org/scenario.html>

²Located in 2025 at <https://www.youtube.com/watch?v=40-5fsVZaII>

³We will explore this tool and try to become familiar with it, but if you would like to train yourself more deeply you could take their online course <https://learn.climateinteractive.org/course/mastering-en-roads>.

- Explain the plot shown in the 10 min En-ROADS introduction video (at 4:54, reproduced below), in which En-ROADS is compared to NGFS. What does the graph show? What is NGFS and why are they comparing to it? What are the three different scenarios plotted? (You might need to look some things up online!)



10.1.1 Identifying the climate science elements of the simulation

In its default setup, the En-ROADS simulation shows a panel of two graphs and a temperature value (click the Home and Reset icons to return to default). Describe these three displays and explain their meanings.

The primary outcome shown, *Greenhouse Gas Net Emissions*, is not a quantity we have explored before. It would be useful to put it into context with other more familiar quantities (like atmospheric CO₂ concentrations and average temperature). Using your own piece of graph paper, make a graph with *four* vertical axes (two on each side) showing these data sets⁴ — Annual CO₂ emissions, Cumulative CO₂ emissions, Annual temperature anomaly, and atmospheric CO₂ (ppm) — and a horizontal axis showing time from 1975 to 2025.

Examining your graphs, explain how our familiar quantities (atmospheric carbon dioxide and global temperature) relate to the emissions annual quantity plotted in En-ROADS.

⁴You should be able to find each of these datasets at ourworldindata.com, but access each individual plot by **Googling** for, e.g., “ourworldindata cumulative co2 emissions” (the ourworldindata internal search engine is bad, so be sure to use google to access these plots). Once at the data graph, remove all country data (“Clear”) and select only the “World” dataset; limit the time range using the slider.

Considering your own graphs, do you see any evidence for the statement⁵ that “temperature is proportional to cumulative emissions”?

What is the implication of this proportionality for the increase in temperature prior to a halt of CO₂ emissions, and the behavior of the temperature after the halt of CO₂ emissions? Discuss this referencing the figures shown below from Dessler Chapter 8.

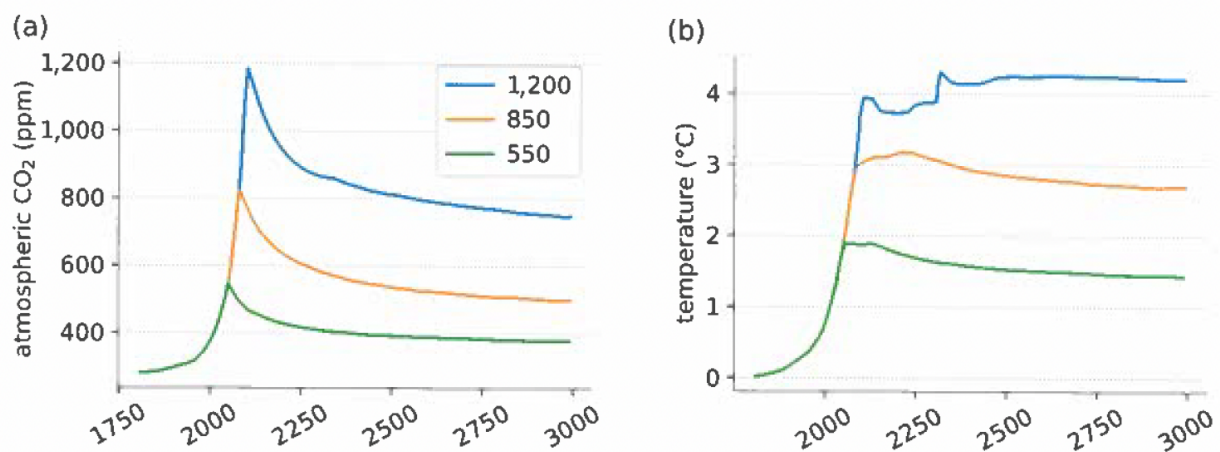


Figure 8.5 (a) Amount of carbon dioxide in the atmosphere as a function of time, for the next 1,000 years. Carbon dioxide emissions rise at 2 percent/year until it hits a peak abundance (550, 850, and 1,200 ppm); then emissions are decreased instantly to zero. (b) The temperature time series corresponding to each carbon dioxide time series. Adapted from Figure 1 of Solomon et al. (2009).

⁵See, e.g., Figure 1b in Jeevanjee et al (AREPS 2025) and the related discussion.

Returning to the simulation, let's examine the climate science inputs to the model and their effects on temperature. There are three such parameters in the model, listed under *Simulation / Assumptions / Earth System / Climate System Sensitivities*. List these quantities (one should be familiar to you, the other two we have not yet discussed) and define them based on information given at the model page and/or your own brief online research.

Now use the *Assumptions* page to adjust these quantities and investigate their impact on the temperature graph. Which has the most influence? If we assume a $\pm 0.5^\circ\text{C}$ uncertainty in the "Climate Sensitivity", how much uncertainty does that lead to in the temperatures at 2050 and 2100? [Use the  button to access the *Temperature Change* graph.]

10.1.2 Societal parameters and choices

This week's reading from Dessler (*Chapter 8*) describes generally how predictive climate models estimate future emissions. This part of a climate model is not part of climate science, but is instead based on societal choices (e.g., about changes in energy sources and usage, and the policies to affect such changes) and societal parameters (e.g., population, economic growth). Review *Section 8.1* from Dessler and then briefly explain:

- The “IPAT” equation⁶ and its constituent parts. What are the meanings and units of each part?
- Why does this equation lead to an estimate of greenhouse-gas emissions? Give some examples.

Within the En-ROADS simulation, the elements of the IPAT equation can be viewed as time-series graphs under *View / Kaya Graphs*. Open these graphs and determine which sliders cause them to be altered. Give a few examples of how the IPAT equation can be changed by such changes, and its effect on *Greenhouse Gas Net Emissions* and *Temperature Change* (returning to views of these graphs). Mention what these changes would mean in the world and/or how they would arise.

⁶This is also known as the *Kaya Identity*, named after Japanese energy economist Yoichi Kaya who developed the approach in the late 20th Century.

Returning to Dessler, review *Section 8.3* and then define **Shared Socioeconomic Pathways**⁷ (SSPs) and briefly describe the five different pathways.

What does, e.g., “SSP1-26” mean? What are the most commonly used pathways? [See Dessler *Figure 8.2* and *Aside 8.3* on naming conventions.]

Return to the En-ROADS simulation and try to mimic the five “pathways” presented in Dessler *Figure 8.2* by adjusting sliders. What types of societal choices and changes lead to each pathway? What is the effect of each on the *Temperature Change* graph?

⁷The SSPs come out of the Intergovernmental Panel on Climate Change (IPCC); more information can be found on [wikipedia](#), and in the [IPCC Synthesis Report](#) (e.g., search the Synthesis Report for SSP).